

method. The data is stored in an inbuilt time series database.

2. **Sensu:** This complements Prometheus. It provides flexibility and scalability to measure the telemetry and execute service checks. The collected data can be more contextual here, and it is possible to extend it to provide automatic remediation workflows as well.
3. **Nagios:** This is the old friend we used to monitor deployments. It serves its purpose well, especially in the context of bare metal deployments to a host of administrators.

Kubernetes monitoring challenges

When we decide to go with Kubernetes, we are ready to embrace the mantra that the only constant in life is change. Naturally, this adds associated challenges with respect to monitoring Kubernetes deployments. Let's discuss a few monitoring challenges to understand them better.

1. The apps are ever evolving; they are always moving.
2. There are many moving pieces to monitor. Kubernetes is not really a monolithic architectural tool, as we all know. Those components also keep on changing.
3. Once upon a time, everything was server based, often bare metal. Then we got cloud based deployment. The natural progression makes multi-cloud a reality, which adds an extra facet to the monitoring challenges.
4. We typically annotate and tag the pods and containers in a Kubernetes deployment to give them nicknames. The monitoring should reflect the same.

Kubernetes data sources

Once we understand the challenges of monitoring in a Kubernetes environment, we need to understand the moving parts of a typical, sizable deployment. These moving parts are the data sources from which Prometheus pulls the monitoring metrics.

1. The hosts (nodes) which are running the Kubelets.
2. The process associated with Kubelets — the Kubelet metrics.
3. Kubelet's built in the cAdvisor data source.
4. The Kubernetes cluster. This is the whole deployment. From a monitoring perspective, these are Kube state metrics.

The good part is that Prometheus can monitor all of the above, addressing all the monitoring challenges we described earlier.

The Prometheus architecture

The heart of Prometheus is its server component, which has a collector that pulls the data from different data sources to store it in an inbuilt time series database (TSDB). It also has an API server (HTTP server) to serve the API request/response.

The unique feature of Prometheus is that it is designed for pulling out data to make it scalable. So, in general, one need not install an agent to the various

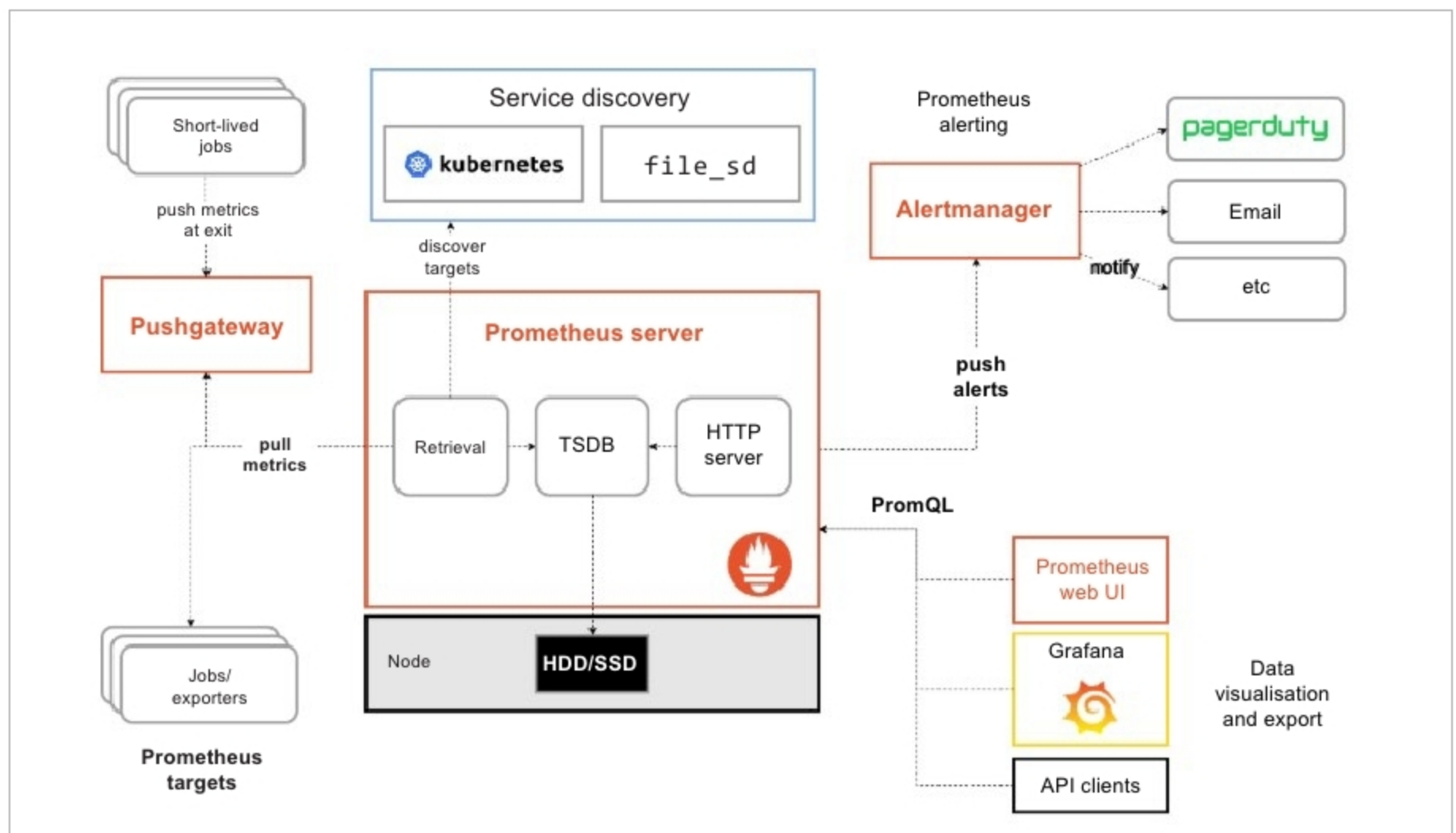


Figure 1: Prometheus architecture (Reference: <https://prometheus.io/docs/introduction/overview/#architecture>)